GENERATIVE AI

Assignment 2

Final Submission

Group 12

Abreu, TIAGO
Álvarez, LUCÍA
Figueras, ALBA
Juncarol, MARTA
Lemos, FRANCISCO

January 21, 2026

Contents

1 Abstract

This project addresses a common challenge faced by researchers: efficiently locating relevant information within large collections of documents. We propose a collaborative retrieval platform that allows users to upload research documents and query specific concepts without manually reviewing all files.

Our approach compares naive and LLM-based chunking strategies and explores metadata extraction at both document and chunk level, integrated using multiple embedding strategies. Retrieval relevance is computed using cosine similarity. Our evaluation shows that a dual representation combining document-level metadata with LLM-based chunking achieves the most robust performance. This final model was integrated into an interactive frontend for practical use.

Code is available at `https://github.com/FranciscoLemos03/GenAI-PR-2025w`

2 Description

2.1 User Journey

This tool allows researchers working on the same project to have a space to share their notes, progress reports, or academic papers, and search for concepts or topics using queries.

Users can access the platform and upload new documents that are relevant to the project in pdf format. These are automatically processed by the system, which divides them in relevant sections and enriches them with semantic metadata. From there, all users from the same group can access their respective documents.

Additionally, users can enter queries to find which documents are closely linked to a topic. This is especially useful with large projects with hundreds of documents.

2.2 Technical Approach

After experimenting with different strategies (as explained in Section 3), these are the methods that provided better results.

2.2.1 Document Segmentation

Documents are divided into chunks to keep the data manageable, as entire files are too large for embedding. We use the *Gemma-3-27b* model to detect subtopics and split the text, as it is a high-capacity free model that provides good results.

2.2.2 Metadata Extraction

To enrich embeddings, metadata is extracted using the *Gemini-2.5-flash* LLM, as it is very efficient for a limited number of requests. Metadata was extracted at both document and chunk levels, but only document-level metadata was finally kept due to better results. The model is prompted to obtain author, year, keywords, topics, and a one-sentence summary, providing a global view of each document.

2.2.3 Embedding Strategy

We used the HuggingFace model *sentence-transformers/all-MiniLM-L6-v2* as the embedder, as it provides sufficient capacity for the task. Embeddings were generated for each chunk and for the metadata, and combined using a weighting factor α. The value of α was optimized during the experiments.

2.2.4 Retrieval

Lastly, semantic retrieval is done through vector similarity search. User queries are embedded with the same model as the documents, and they are compared against all embeddings through cosine similarity.

2.2.5 Database

Uploaded documents are stored in a data folder, while all metadata and embeddings are saved in a *database.json* file, allowing retrieval without recomputation.

2.2.6 Web Application

The program can be run either directly from the terminal via *backend/main.py* or through a web application server. Setup instructions are provided in the *Project_ Setup_ Guide_ 12* manual.

2.3 Challenges

One main challenges was the initial scope of the project, which was highly ambitious and other functionalities, such as automatic summarization. Due to time and resource constraints, we had to narrow the scope and focus on building a robust retrieval system.

A second major challenge was integrating the LLM for text chunking. Since we relied on free-tier models, computational resources were limited (for example, requests per day), which required exploring alternative strategies.

3 Evaluation

For the evaluation, we define a fictitious scenario in which four researchers collaborate on a multimodal AI system for the early diagnosis of Alzheimer's disease. The dataset consists of 20 arXiv papers (covering ML, AI, Alzheimer's research, and teamwork) and 8 synthetic documents. The synthetic documents are generated using different LLMs to simulate distinct writing styles, representing progress reports and research notes from each researcher.

Besides, we created query-document pairs to compare retrieval under different levels of difficulty, and using single or multi-document categories. Retrieval is measured using the following metrics:

- Mean Reciprocal Rank (MRR)
- Hit Rate@3
- Mean Average Precision (MAP) on multi-document queries.

In addition, a separate evaluation was conducted using adversarial trap queries to analyze the robustness.

The notebooks for all experiments can be found in the *backend/experiments* folder.

3.1 Experiment 1: Chunking

Two chunking approaches were compared:

- **Naive chunking**: splitting at a predefined number of tokens.
- **LLM-based chunking**: using *Gemma-3-27b* to detect subtopics for splitting.

While naive chunking shows a slight advantage in some metrics (Figure 1), Figure 2 indicates that LLM-based chunking is more robust for trap queries. Since the overall score differences are small, we keep LLM-based chunking for the next experiments.

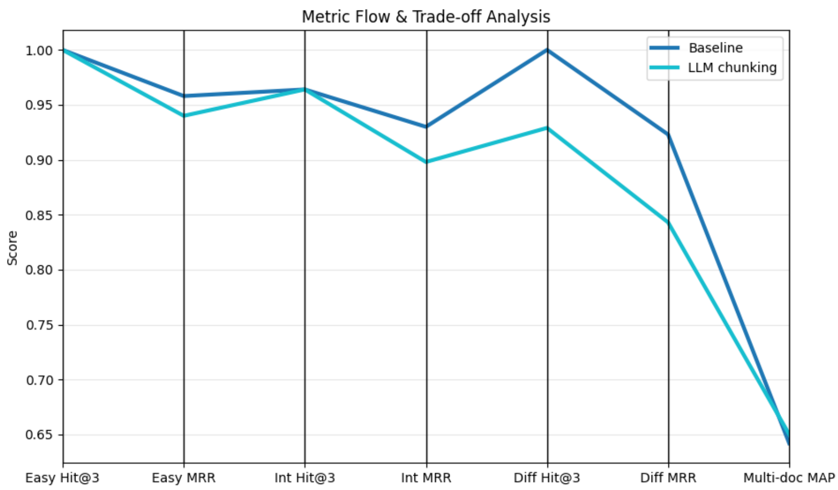

Figure 1: Results of different metrics for Experiment 1

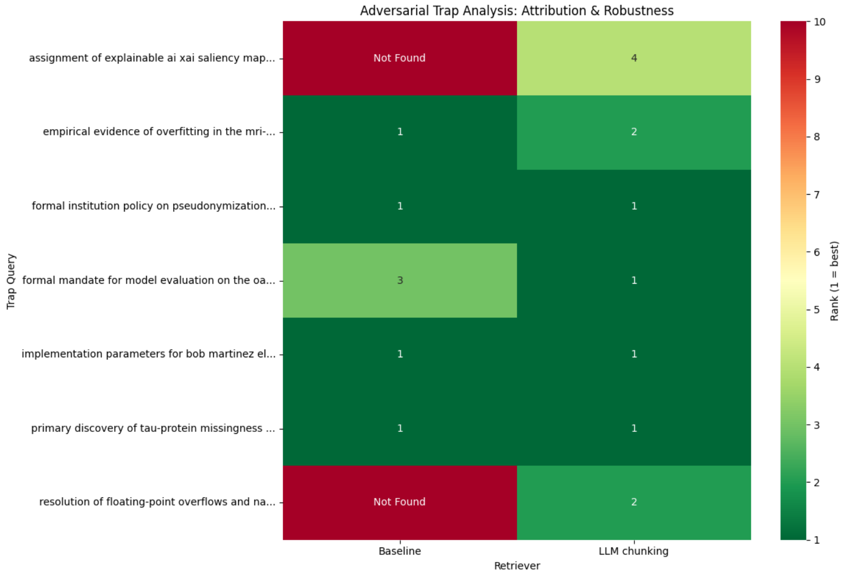

Figure 2: Adversarial Trap Heatmap for Experiment 1

3.2 Experiment 2: Document-level metadata

This experiment extracts document-level metadata and compares different strategies to incorporate it into the raw text:

- **Baseline**: embeddings computed using only the raw text.
- **Metadata-only**: embeddings computed using only metadata.
- **Concatenation**: embeddings generated after concatenating chunk text with its corresponding metadata.
- **Scoring / dual**: metadata and raw text are embedded separately, and combined using a weighting score α.

The results indicate that the dual-embedding approach performs best, achieving the highest retrieval metrics (Figure 3) and demonstrating strong robustness against adversarial traps (Figure 4).

The dual-embedding uses the parameter α, which controls the relative weighting of chunk and metadata embeddings. For this, we run another experiment and find out that the best performance was achieved with $\alpha = 0.5$, assigning equal importance to both embeddings, as shown in Figure 5.

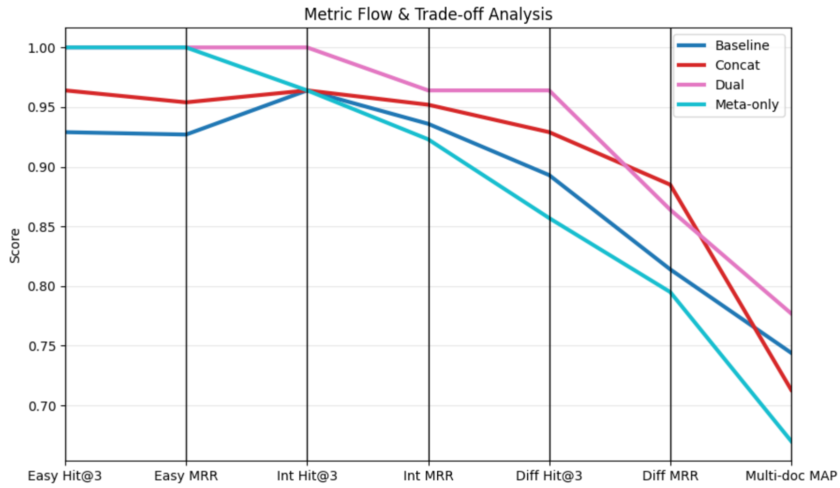

Figure 3: Results of different metrics for Experiment 2

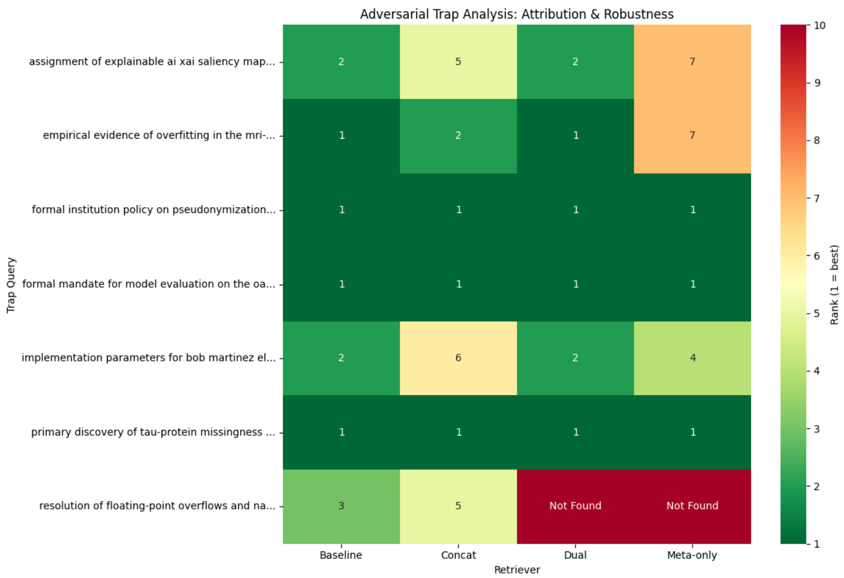

Figure 4: Adversarial Trap Heatmap for Experiment 2

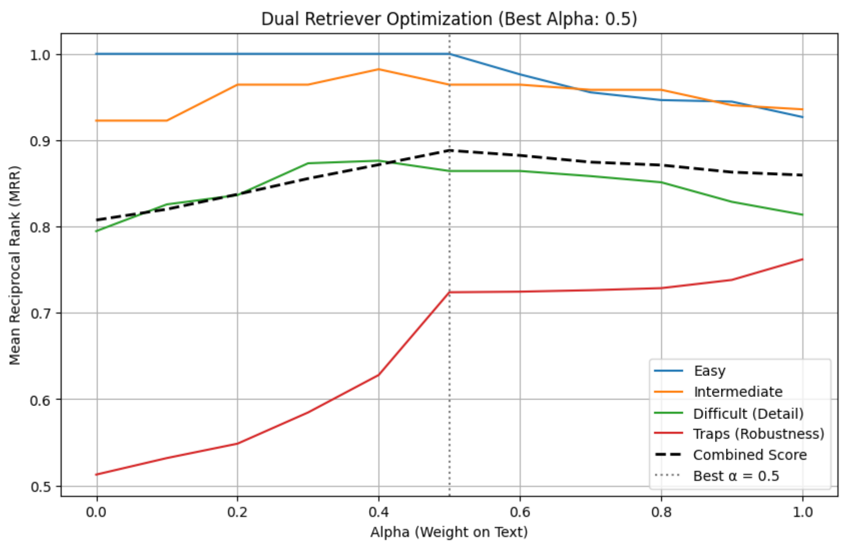

Figure 5: Alpha parameter optimization

3.3 Experiment 3: Chunk-level metadata

In this experiment, we evaluate the impact of creating chunk-level metadata, using the same four incorporation techniques as before. The chunk metadata includes the main concept, its type (background, method, result, etc.), and a small set of keywords. This aims to capture semantic details that may be missed by document-level metadata alone. However, as shown in Figure 6 and Figure 7, the baseline model without chunk-level metadata provided the best results.

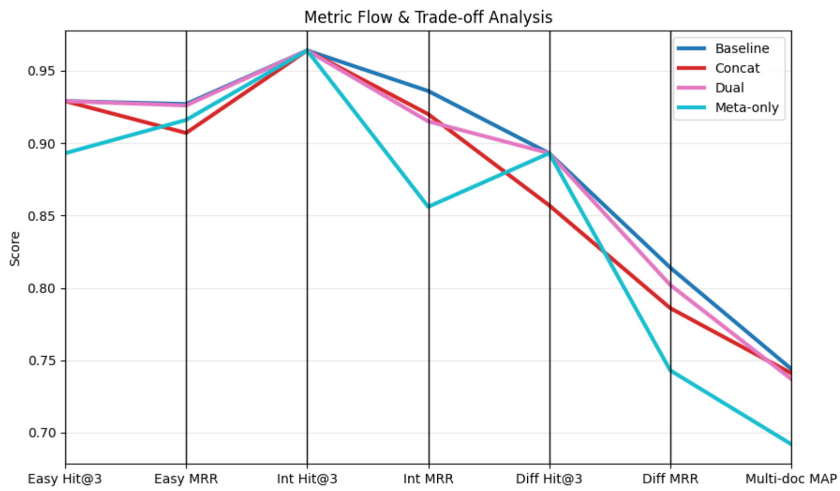

Figure 6: Results of different metrics for Experiment 3

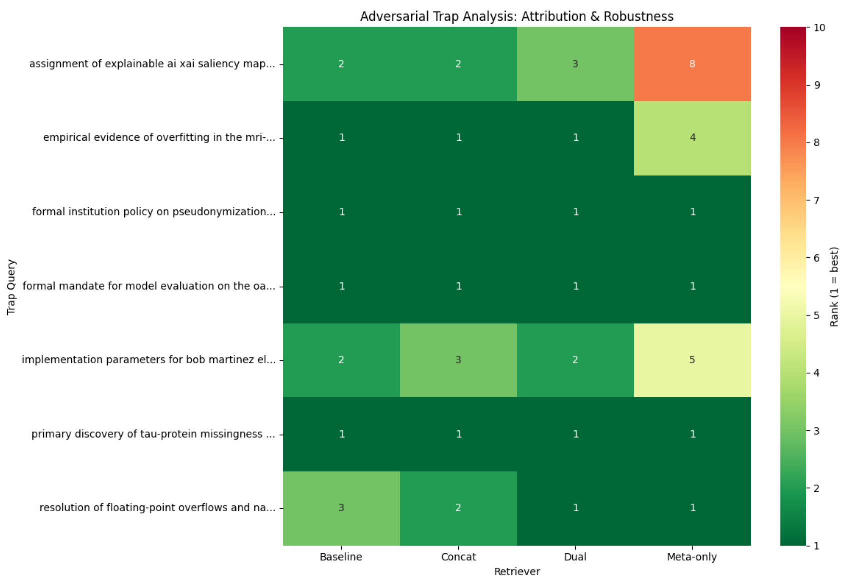

Figure 7: Adversarial Trap Heatmap for Experiment 3

3.4 Experiment 4: Final Model Optimization

Based on the results of the previous experiments, we selected LLM-based chunking with whole-text metadata and a dual-embedding integration strategy. To further improve performance, we tuned the cosine similarity threshold for retrieval using also negative queries, and found the optimal value to be 0.45, as shown in Figure 8.

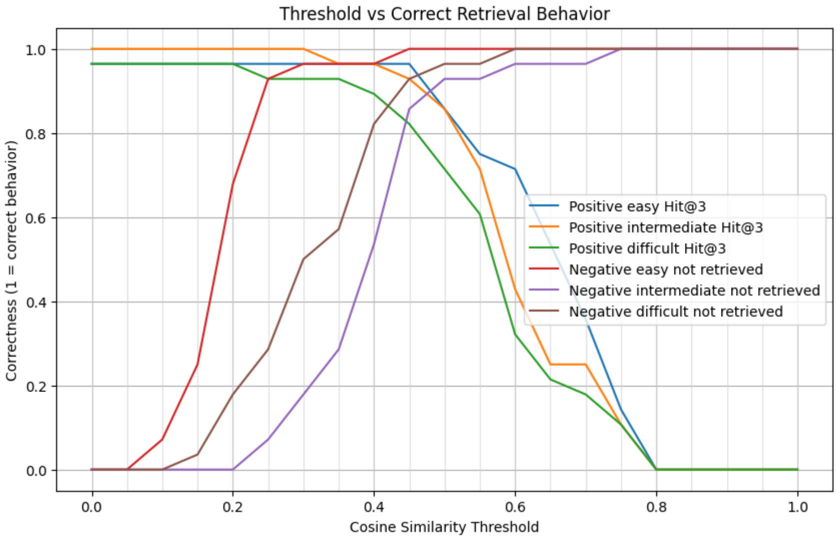

Figure 8: Threshold parameter optimization

We integrated this final model with a web application.

4 Reflection

Our system increases human agency, as it supports researchers by making document retrieval easier without taking decisions on their behalf. The system provides enough information for the users to understand why a document was retrieved, and to judge the relevance of the results themselves.

The project was valuable for understanding LLM-based systems. We would consider further development, as it could be valuable for collaborative research environments, and be extended with additional features.